Multi-Channel based Light Inception with boosTing tEchniques for Time Series Classification

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***Abstract*—Time series classification (TSC) is a fundamental technology for anomaly detection, prediction, and identification, and its application covers a wide range of fields including healthcare, human activity recognition, cybersecurity, and finance. Although there are a number of deep learning–based models that can accurately classify time series data, it is important to reduce the size of the model while maintaining accuracy for its implementation on edge devices, such as Internet of Things (IoT) devices. This allows for the real-time processing and reduces power consumption. In this paper, we propose two lightweight models: the Multi-Channel MACD-Histogram-based LITE (MC-MHLITE) model and its ensemble variant, the Parallel Multi-Channel MACD-Histogram-based LITE (PMC-MHLITE) model. In the MC-MHLITE model, MACD histogram is integrated into the Light Inception with boosTing tEchniques (LITE) model to enhance feature extraction. The PMC-MHLITE model is designed as a parallel architecture comprising multiple MC-MHLITE units to further improve the versatility of the classification performance. These models aim to improve accuracy by incorporating diverse features while maintaining the efficiency of the LITE model.**

***Index Terms*—Time series classification, Deep learning, the Light Inception with boosTing tEchniques model, MACD histogram**

I. Introduction

In recent years, time series data analysis has received significant attention across a wide range of fields, including healthcare, human activity recognition, cybersecurity, finance, and marketing [1]. In time series data analysis, the task of assigning a class label to unlabeled time series data is referred to as time series classification (TSC) [2]. TSC is a fundamental technology for anomaly detection, prediction, and classification. Consequently, the research on developing its core, high-precision classification models is a significant challenge. In particular, time series data have their own complexities such as variations in sampling intervals, noise interference, and phase shifts over time. These characteristics make it difficult to directly apply general-purpose data classification methods to TSC.

Therefore, many classification models based on deep learning have been proposed for TSC [3]. In particular, convolutional neural networks (CNNs) [4] are an effective technique; one such representative example is the Fully Convolutional Network (FCN) model [5]. The FCN model efficiently extracts both local and global features from time series data using stacked one-dimensional (1D) CNNs. In addition, the LSTM-FCN [6] model, which combines Long Short-Term Memory (LSTM) [7] with the FCN model, has also been proposed. These ensemble approaches achieve higher classification accuracy than conventional machine learning techniques.

Subsequently, as successors to these models, the Multi-Channel MACD-Histogram LSTM-FCN (MC-MHLF) model utilizing the Moving Average Convergence/Divergence (MACD) histogram [8], and its variants, the MC2-MHLF and PMC2-MHLF models [9], have been proposed. These models calculate the MACD histogram of time series data, which represents the velocity and acceleration of the data, and treat the time series data and MACD histograms as multi-channel data. This emphasizes inflection points in time series data increases and decreases, enabling improved classification accuracy.

While these models offer high classification accuracy, the number of parameters in these models has increased; however, lightweight models are required for real-world applications due to real-time processing and the resource constraints of sensor and edge devices. To meet this requirement, research has focused on efficient models, including model compression via knowledge distillation or compact architectures, such as Light Inception with boosTing tEchniques (LITE) [10], [11], which utilizes depthwise separable convolutions.

In this paper, we propose a novel method that integrates the LITE model with the concept of the MC-MHLF model to achieve both lightweight and the superior feature extraction capabilities of the MC-MHLF model. The main contributions of this study are as follows:

- We propose the Multi-Channel MACD-Histogram-based LITE (MC-MHLITE) model and its ensemble variant, the Parallel-Multi-Channel MACD-Histogram-based LITE (PMC-MHLITE) model. The MC-MHLITE model integrates both long-term and short-term MACD histograms into the input layer of the LITE model. The PMC-MHLITE model is a parallel architecture consisting of multiple MC-MHLITE units. Each unit receives input from multi-channel data created using long- and short-term parameters, respectively.

- We evaluated the new models using the UCR time series archive [12]. In addition to classification accuracy, a comprehensive performance evaluation was conducted using the number of model parameters and processing time as criteria.

The remainder of this paper is organized as follows. Section II discusses related work on lightweighting. Section III provides background knowledge of TSC, MACD histogram, and the MC-MHLF model. Section IV proposes new models. Section V reports the results of experiments. Finally, Section VI concludes the paper.

II. Related Work

With the acceleration of deep learning applications in the real world, deploying models on edge devices with constrained computational resources has become a critical challenge. Compared to the high-performance backend environments such as servers and workstations used for model training, edge devices operate under severe power, computation, and memory constraints. Therefore, it is essential to build lightweight models that can operate under restricted computational environments. Therefore, a significant research area has emerged: reducing the size of models while maintaining the performance of large-scale, high-precision models built on the backend.

In the field of model lightweighting, four primary approaches are widely recognized: pruning [13], quantization [14], knowledge distillation [15], and network architecture optimization.

Pruning is a technique used to eliminate redundant parameters within a neural network. Proposed methods commonly include removing connections with small absolute weights or pruning at the filter level. Quantization is a technique that reduces computational cost by converting model parameters from standard 64-bit or 32-bit floating-point numbers to lower-precision integers. Knowledge distillation is a technique that trains a small-scale model (the student model) to mimic the output of a pre-trained, high-accuracy, large-scale model (the teacher model).

Furthermore, approaches that fundamentally redesign the model architecture itself have gained significant attention. For instance, MobileNets [16] achieved a substantial reduction in computational complexity by introducing depthwise separable convolution [17]. ShuffleNet [18] combines channel shuffling with group convolutions to optimize the architecture for edge devices.

Although model architecture optimization was initially developed for image data, research on lightweight models is also advancing in the field of TSC. For example, in the paper [19], a novel model compression learning framework utilizing knowledge distillation was proposed for time series data classification models. Furthermore, the LITE model [10], [11] employs an Inception-like architecture, which incorporates parallel convolutional layers with various kernel sizes. This model achieves a significant reduction in both computational complexity and the number of parameters by employing depthwise separable convolutions instead of standard convolutions.

III. Preliminaries

A. TSC

A time series is defined as a sequence of elements recorded in time order. Let x_t denote the element at time t; a univariate time series $\mathbf{x}$ of length n is represented as follows:

$$\mathbf{x} = \langle x_1, x_2, \ldots, x_n \rangle, \quad x_t \in \mathbb{R}$$

For multivariate time series, x_t is defined as an M-dimensional vector $x_t \in \mathbb{R}^M$, and the entire time series is described as a matrix $\mathbf{x} \in \mathbb{R}^{M \times n}$.

TSC is defined as the task of predicting the class label for a given time series. Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote the training dataset, where $\mathbf{x}_i$ represents the i-th time series, defined as $\mathbf{x}_i = \langle x_{i,1}, x_{i,2}, \ldots, x_{i,n} \rangle$. The corresponding class label is given by $y_i \in \mathcal{Y} = \{1, 2, \ldots, C\}$, where n is the length of the time series and C is the total number of classes. The goal of TSC is to construct a classification model $f : \mathcal{X} \to \mathcal{Y}$ that predicts the class label $y = f(\mathbf{x})$ for an unlabeled time series $\mathbf{x}$.

B. MACD Histogram

The MACD histogram [8] is an indicator originally developed for technical analysis in financial markets, calculated using exponential moving averages (EMA). For a given time series $\mathbf{x}$, the i-th element of EMA with a window size $\omega \in \mathbb{N}$ is defined as follows:

$$ema(\omega)[\mathbf{x}]_i = \alpha \cdot x_i + (1 - \alpha) \cdot ema(\omega)[\mathbf{x}]_{i-1} \tag{1}$$

Note that x_i represents the i-th value of $\mathbf{x}$, and $\alpha = 2/(\omega + 1)$ is the smoothing coefficient. During the initial period where $i \leq \omega$, the EMA is calculated as the simple arithmetic mean as follows:

$$ema(\omega)[\mathbf{x}]_i = \frac{1}{i} \sum_{k=1}^{i} x_k \tag{2}$$

The MACD line, which represents the momentum of time series data, is calculated as the difference between two EMAs with distinct window sizes, ω_1 and ω_2 (where $\omega_1 < \omega_2$):

$$macd(\omega_1, \omega_2)[\mathbf{x}]_i = ema(\omega_1)[\mathbf{x}]_i - ema(\omega_2)[\mathbf{x}]_i \tag{3}$$

Furthermore, the Signal line is generated by applying an EMA with a window size ω_3 to the MACD values:

$$signal(\omega_1, \omega_2, \omega_3)[\mathbf{x}]_i = ema(\omega_3)[macd(\omega_1, \omega_2)[\mathbf{x}]]_i \tag{4}$$

The MACD histogram is defined as the difference between the MACD and Signal lines, serving as an indicator that captures the acceleration of changes in the data.

$$\begin{aligned} hist(\omega_1, \omega_2, \omega_3)[\mathbf{x}]_i = {} & macd(\omega_1, \omega_2)[\mathbf{x}]_i \\ & - signal(\omega_1, \omega_2, \omega_3)[\mathbf{x}]_i \end{aligned} \tag{5}$$

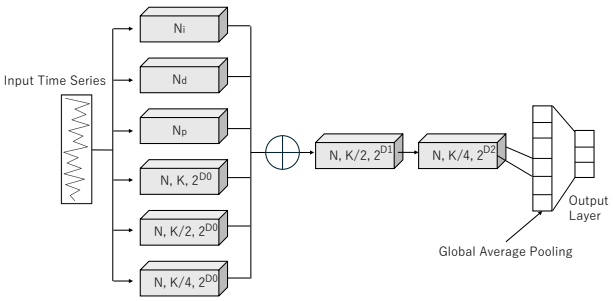


Fig. 1: LITE Model

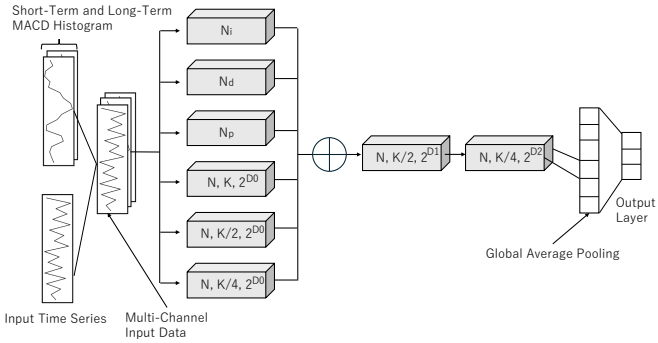


Fig. 2: MC-MHLITE Model

C. MC-MHLF Model

The MC-MHLF model combines long-term and short-term MACD histograms calculated from time series data and uses them as input as three-dimensional multichannel data. The model has a parallel architecture that consists of an LSTM unit and an FCN unit. The features extracted from both units are fused in a concatenation layer before being propagated to the output layer. The FCN consists of three convolutional layers, followed by a global average pooling (GAP) layer. To stabilize the training process and introduce nonlinearity, batch normalization and the rectified linear unit (ReLU) activation function are applied to each convolutional layer. Finally, the GAP layer extracts representative values from the feature maps as feature values and aggregates the extracted feature values into a single feature vector.

D. LITE Model

The LITE model is a lightweight version of InceptionTime that significantly reduces the number of model parameters. To achieve this reduction while maintaining performance, it employs boosting techniques such as standard convolutions and custom filters, in addition to depth-wise separable convolutions.

IV. Proposed Model

In this paper, we propose the modified version of LITE model; MC-MHLITE and PMC-MHLITE models.

A. MC-MHLITE Model

Figure 2 shows the structure of MC-MHLITE Model. In this model, there are three types of input data: the original time series data, and the short- and long-term MACD histograms. In this model, these data are composed as 3-dimensional multi-channel data. For the time series data $\mathbf{x}$, let the MACD histogram calculated using long-term parameters be denoted as $hist(\omega_{l_1}, \omega_{l_2}, \omega_{l_3})[\mathbf{x}]$, and the MACD histogram calculated using short-term parameters as $hist(\omega_{s_1}, \omega_{s_2}, \omega_{s_3})[\mathbf{x}]$. Here, the ith element of the 3-dimensional multi-channel data is

$$multi3(\mathbf{x})_j = \langle x_i, hist(\omega_{l_1}, \omega_{l_2}, \omega_{l_3})[\mathbf{x}]_i, \\ hist(\omega_{s_1}, \omega_{s_2}, \omega_{s_3})[\mathbf{x}]_i \rangle. \quad (6)$$

In this study, the parameters $\omega_{s_1} = 3,\ \omega_{s_2} = 5,\ and\ \omega_{s_3} = 4$ are used to extract the short-term MACD and MACD histogram from the time series data. In addition, the long-term MACD and MACD histogram of time series data is extracted by using $\omega_{l_1} = 13,\ \omega_{l_2} = 26,\ and\ \omega_{l_3} = 9$.

By incorporating the MACD histogram—which represents the acceleration of the time series data—into a multi-channel framework, weights are assigned to the features. Furthermore, the model emphasizes not only changes in the values of the time series data itself but also the simultaneous incorporation of the "acceleration" and trend changes of the time series data as represented by the MACD histograms. While the LITE model achieves lightweight performance by using depth-wise separable convolutions, accepting 3-dimensional multi-channel data as input enables it to learn both long-term fluctuation trends—which cannot be fully captured by a single time series—and short-term changes simultaneously. This emphasizes characteristic sub-time series and their changes, leading to an expected improvement in classification accuracy.

B. PMC-MHLITE Model

Figure 3 shows the structure of PMC-MHLITE Model. This model consists of two LITE models running in parallel: one that takes time series data and a long-term MACD histogram as inputs, and another that takes time series data and a short-term MACD histogram as inputs. For the time series data $\mathbf{x}$, let the MACD histogram calculated using long-term parameters be denoted as $hist(\omega_{l_1}, \omega_{l_2}, \omega_{l_3})[\mathbf{x}]$, and the MACD histogram calculated using short-term parameters as $hist(\omega_{s_1}, \omega_{s_2}, \omega_{s_3})[\mathbf{x}]$. Here, the ith element of the two multi-channel data sets input to PMC-MHLITE is

$$lmulti2(\mathbf{x})_j = \langle x_i, hist(\omega_{l_1}, \omega_{l_2}, \omega_{l_3})[\mathbf{x}]_i \rangle \quad (7)$$

$$smulti2(\mathbf{x})_j = \langle x_i, hist(\omega_{s_1}, \omega_{s_2}, \omega_{s_3})[\mathbf{x}]_i \rangle \quad (8)$$

The outputs from the Global Average Pooling layers of these LITE models are combined in a fusion layer and fed into the output layer. By combining the long-term and short-term MACD histograms with the time series data to create a multi-channel input, the aim is to process different sub-time series data using separate LITE models, thereby extracting

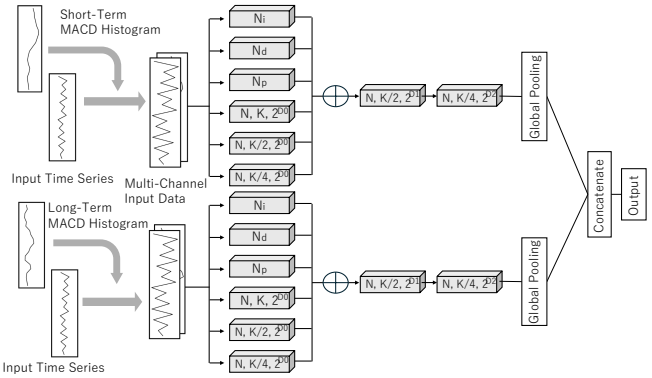


Fig. 3: PMC-MHLITE Model

features from each dataset without compromising their distinct characteristics. Furthermore, since the LITE model itself is a lightweight model with a reduced number of parameters, even when two models are deployed in parallel, it is believed that high-precision classification utilizing more features is possible while keeping the increase in computational cost to a minimum.

V. Experiments

A. Dataset

In the experiments, we use the UCR Time Series Classification Archive released by the University of California, Riverside, which is widely used as a benchmark for time series classification. This archive consists of 85 datasets; each dataset contains time series data with varying lengths and has been pre-split into training and test datasets.

These datasets are classified into the following seven categories based on their characteristics.

- **Device**: Time series representing the states or usage patterns of electronic appliances and devices.
- **ECG**: Biometric signals captured via electrocardiography (ECG) to monitor cardiac activity.
- **Image**: Time series converted from image outlines or shapes.
- **Motion**: Data recording the physical movements and trajectories of human subjects or objects.
- **Sensor**: Physical measurements observed by environmental or industrial sensors.
- **Simulated**: Synthetic datasets generated through mathematical models or computational simulations.
- **Spectro**: Spectral data of substances (e.g., food or chemical compounds) acquired using a spectrometer.

B. Criteria

For each dataset, we trained the model using the training dataset and measured the accuracy using the test dataset. In the experiment, we compared classification accuracy, the number of parameters, and processing time with other models. The comparison models included the conventional MC-MHLF model and the LITE model.

TABLE I: MHLF's parameters

	Layer-1	Layer-2	Layer-3
Number of Filters	128	256	128
Size of Kernel	8	5	3

TABLE II: Summary of Comparison Results by Category

Category	Metric	LITE	MC-MHLITE	PMC-MHLITE
Device (6)	Average of Accuracy	**0.735**	0.728	0.725
	Average of Ranking	**1.500**	2.167	2.333
	Number of Best Accuracy	**4**	1	1
ECG (6)	Average of Accuracy	0.960	**0.963**	0.961
	Average of Ranking	**1.667**	1.833	**1.667**
	Number of Best Accuracy	**3**	2	**3**
Image (29)	Average of Accuracy	0.859	0.866	**0.869**
	Average of Ranking	2.172	**1.862**	**1.862**
	Number of Best Accuracy	10	10	**11**
Motion (14)	Average of Accuracy	0.783	0.788	**0.795**
	Average of Ranking	2.214	1.786	**1.643**
	Number of Best Accuracy	4	5	**7**
Sensor (18)	Average of Accuracy	0.862	0.861	**0.869**
	Average of Ranking	1.778	2.056	**1.333**
	Number of Best Accuracy	9	6	**13**
Simulated (5)	Average of Accuracy	0.965	0.971	**0.989**
	Average of Ranking	**1.400**	2.000	**1.400**
	Number of Best Accuracy	**4**	2	3
Spectro (7)	Average of Accuracy	**0.900**	0.886	0.897
	Average of Ranking	1.571	2.286	**1.429**
	Number of Best Accuracy	**5**	1	**5**
Total Average	Average of Accuracy	0.855	0.857	**0.863**
	Average of Ranking	1.941	1.988	**1.671**
	Number of Best Accuracy	38	26	**43**

We evaluated the models using three metrics across all 85 datasets: average accuracy, average of ranking, and number of best accuracy. The average accuracy is the mean accuracy achieved for the test data across all 85 datasets. The average of ranking is calculated by determining the ranking of the accuracy rates among the models for each dataset, and then averaging these rankings across the 85 datasets. The number of best accuracy is the number of datasets in which a model achieved the highest accuracy rate, after determining the ranking of accuracy rates among models for each dataset.

C. Environment

Table I shows the parameters for each convolutional layer of the FCN and the number of units in LSTM was set to 128. The LITE architecture setup employs $N_i = 6$ and $N_d = 6$ (kernel sizes: $[2, 4, 8, 16, 32, 64]$), $N_p = 5$ (kernel sizes: $[6, 12, 24, 48, 96]$), $N = 32$, and $K = 40$. Dilation rates are initialized at $D_0 = 1$ and incremented with network depth to $D_1 = 2$ and $D_2 = 4$. Processing time was measured using a computer equipped with an NVIDIA GeForce RTX 4070 Ti (CPU: Intel(R) Core(TM) i7-4790 @ 3.60GHz, Memory: DDR3-1600 (PC3-12800) 16GB, Storage: 1TB WD Blue 3D NAND SATA WDS100T2B0A).

D. Experiment 1

Table II shows the results of a performance comparison between the LITE model, the MC-MHLITE model, and the PMC-MHLITE model across seven categories. Overall, the PMC-MHLITE model showed the best performance.The PMC-MHLITE model ranked first in all metrics, with the

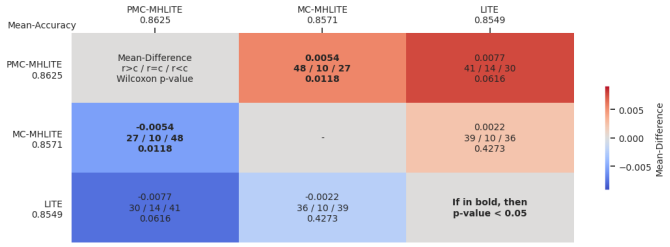


Fig. 4: Multi Comparison Matrix

TABLE III: Summary of Comparison Results by Category (Including MC-MHLF)

Category	Metric	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
Device (6)	Average of Accuracy	**0.755**	0.735	0.728	0.725
	Average of Ranking	**1.167**	2.333	3.167	3.333
	Number of Best Accuracy	**5**	1	0	0
ECG (6)	Average of Accuracy	**0.968**	0.960	0.963	0.961
	Average of Ranking	**1.000**	2.333	2.500	2.500
	Number of Best Accuracy	**6**	2	2	1
Image (29)	Average of Accuracy	**0.887**	0.859	0.866	0.869
	Average of Ranking	**1.379**	3.034	2.586	2.724
	Number of Best Accuracy	**22**	4	3	2
Motion (14)	Average of Accuracy	**0.806**	0.783	0.788	0.795
	Average of Ranking	**1.857**	2.857	2.429	2.071
	Number of Best Accuracy	**7**	4	3	4
Sensor (18)	Average of Accuracy	**0.878**	0.862	0.861	0.869
	Average of Ranking	2.000	2.222	2.556	**1.833**
	Number of Best Accuracy	**11**	6	4	9
Simulated (5)	Average of Accuracy	**0.991**	0.965	0.971	0.989
	Average of Ranking	2.000	1.800	2.600	**1.600**
	Number of Best Accuracy	**3**	**3**	2	**3**
Spectro (7)	Average of Accuracy	**0.941**	0.900	0.886	0.897
	Average of Ranking	**1.286**	2.000	3.143	2.000
	Number of Best Accuracy	**5**	4	1	3
Total Average (85)	Average of Accuracy	**0.879**	0.855	0.857	0.863
	Average of Ranking	**1.576**	2.612	2.682	2.341
	Number of Best Accuracy	**59**	23	14	21
	Processing Time [s]	1895.194	**995.820**	1020.266	1421.505
	Number of Parameters	273,619	**10,859**	16,215	27,067

average of accuracy of 0.863, the average of ranking of 1.671, and the number of best accuracy of 43. In particular, it performs well in categories with a large number of datasets, such as "Image," "Motion," and "Sensor." In contrast, the LITE model is superior in a specific category. In particular, in "Device" category, the LITE model outperforms others with the average of ranking of 1.500 and the number of best accuracy of 4. This is because the Device category has data that is similar to spike signals or digital signals; therefore, it would be counter-productive to add a MACD histogram. The MC-MHLITE model showed good accuracy only in "ECG," but the number of best accuracy was achieved in only 26 datasets. The average rank was also the lowest. In conclusion, the PMC-MHLITE model is the best for a wide range of datasets. However, it is also necessary to select a model according to the characteristics of the target category.

E. Evaluation Experiment 2

Table III provides a comprehensive comparison of the performance of each model across all 85 datasets. We discuss the trade-off between the effectiveness and computational cost of the proposed model based on these results.

First, in terms of accuracy, the MC-MHLF model recorded an average correct rate of 0.879 and an average of ranking of 1.576, demonstrating the best performance among the four models. However, the MC-MHLF model has a notably high number of parameters—approximately 274,000—and the longest average processing time at 1,895.194 seconds. In contrast, the MC-MHLITE model, which was designed to be more lightweight, reduced the number of parameters to approximately one-seventeenth (16,215) of that of the MC-MHLF model, while still achieving an average accuracy of 0.857, surpassing the baseline LITE model (0.855).

Furthermore, focusing on the PMC-MHLITE model, the average accuracy improved to 0.863, and the average of ranking also improved to 2.341. Although the number of parameters has increased slightly to 27,066, it remains about one-tenth the size of the MC-MHLF model. In terms of processing time, while the MC-MHLF model takes approximately 1,900 seconds, the PMC-MHLITE model takes about 1,422 seconds, achieving a reduction of approximately 25%.

VI. Conclusion

In this paper, we proposed the MC-MHLITE and PMC-MHLITE models to achieve both high accuracy and lightweight performance for TSC. These models integrate MACD histograms, which capture dynamic changes in time series, into the LITE model. By introducing short- and long-term MACD histograms that represent the acceleration of time series data, we enabled the efficient learning of trend changes that are difficult to extract from a single time series alone. The experimental results show that the PMC-MHLITE model is the best overall performance compared with the LITE and MC-MHLITE models. Compared to the MC-MHLF model, the PMC-MHLITE model is about one-tenth the size in parameters and 25% faster in processing time. In our future work, we will develop more accurate lightweight models based on our approaches. We succeeded in reducing parameters, but the effect of processing time reduction was small due to library problems in implementation. We are going to address implementation issues and speed up the process.

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Appendix A
Detailed Comparison of Classification Accuracy

This appendix provides a comprehensive evaluation of classification accuracy across 85 datasets from the UCR Time Series Archive, categorized into seven major domains as shown in Table IV (Device), Table V (ECG), Table VI (Image), Table VII (Motion), Table VIII (Sensor), Table IX (Simulated), and Table X (Spectro). The performance of four models: MC-MHLF, LITE, MC-MHLITE, and PMC-MHLITE models, is compared based on average accuracy, average ranking, and the number of best accuracy results.

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TABLE IV: Comparison of classification accuracy for each model in the "Device" category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
Computers	0.816	**0.824**	0.784	0.792
ElectricDevices	**0.761**	0.714	0.707	0.712
LargeKitchen Appliances	**0.912**	0.891	0.888	0.877
Refrigeration Devices	**0.589**	0.576	0.579	0.587
ScreenType	**0.621**	0.603	0.616	0.595
SmallKitchen Appliances	**0.832**	0.800	0.792	0.784
Average of Accuracy	**0.755**	0.735	0.728	0.724
Average of Ranking	**1.167**	2.333	3.167	3.333
Number of Best Accuracy	**5**	1	0	0

TABLE V: Comparison of classification accuracy for each model in the "ECG" category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
ECG200	**0.930**	0.910	**0.930**	0.910
ECG5000	**0.948**	**0.948**	0.946	0.945
ECGFiveDays	**1.000**	**1.000**	**1.000**	**1.000**
NonInvasiveFatal ECG_Thorax1	**0.962**	0.950	0.957	0.960
NonInvasiveFatal ECG_Thorax2	**0.965**	0.955	0.950	0.953
TwoLeadECG	**1.000**	0.997	0.997	0.999
Average of Accuracy	**0.968**	0.960	0.963	0.961
Average of Ranking	**1.000**	2.333	2.500	2.500
Number of Best Accuracy	**6**	2	2	1

TABLE VI: Comparison of classification accuracy for each model in the "Image" category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
50words	**0.820**	0.745	0.787	0.778
Adiac	**0.867**	0.801	0.818	0.857
ArrowHead	**0.909**	0.777	0.823	0.829
BeetleFly	**1.000**	0.950	**1.000**	0.950
BirdChicken	0.950	**1.000**	0.950	**1.000**
DiatomSize Reduction	**0.980**	0.958	0.931	0.961
DistalPhalanx OutlineAgeGroup	**0.863**	0.835	0.845	0.808
DistalPhalanx OutlineCorrect	**0.843**	0.818	0.812	0.817
DistalPhalanxTW	**0.808**	0.775	0.793	0.790
FaceAll	**0.968**	0.856	0.921	0.927
FaceFour	**0.989**	0.955	0.920	0.943
FacesUCR	0.961	0.960	**0.963**	0.961
FISH	0.994	0.989	0.994	**1.000**
HandOutlines	0.883	**0.914**	0.892	0.890
Herring	0.750	**0.766**	0.750	0.734
MedicalImages	**0.782**	0.775	0.758	0.771
MiddlePhalanx OutlineAgeGroup	**0.813**	0.788	0.783	0.775
MiddlePhalanx OutlineCorrect	**0.848**	0.810	0.838	0.822
MiddlePhalanxTW	**0.647**	0.639	0.627	0.629
OSULeaf	**1.000**	0.950	0.996	0.992
Phalanges OutlinesCorrect	**0.848**	0.843	0.845	0.846
ProximalPhalanx OutlineAgeGroup	0.883	0.878	**0.888**	0.873
ProximalPhalanx OutlineCorrect	**0.931**	0.921	0.914	0.911
ProximalPhalanxTW	0.817	**0.827**	0.825	0.815
ShapesAll	**0.943**	0.888	0.900	0.937
SwedishLeaf	**0.989**	0.960	0.971	0.968
Symbols	**0.981**	0.963	0.972	0.972
WordsSynonyms	**0.724**	0.655	0.683	0.716
yoga	**0.935**	0.907	0.912	0.924
Average of Accuracy	**0.887**	0.859	0.866	0.869
Average of Ranking	**1.379**	3.034	2.586	2.724
Number of Best Accuracy	**22**	4	3	2

TABLE VII: Comparison of classification accuracy for each model in the "Motion" category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
Cricket_X	0.815	**0.821**	0.787	0.818
Cricket_Y	0.831	**0.862**	0.815	0.831
Cricket_Z	0.869	0.841	0.841	**0.877**
Gun_Point	**1.000**	**1.000**	**1.000**	**1.000**
Haptics	0.591	0.565	**0.604**	0.591
InlineSkate	**0.487**	0.422	0.465	0.471
ToeSegmentation1	**0.982**	**0.982**	0.969	0.965
ToeSegmentation2	**0.962**	0.923	0.938	0.915
uWaveGesture Library_X	0.819	0.812	**0.833**	0.831
uWaveGesture Library_Y	0.733	0.716	0.741	**0.743**
uWaveGesture Library_Z	**0.775**	0.756	0.772	0.764
uWaveGesture LibraryAll	0.895	0.900	0.905	**0.911**
Worms	**0.735**	0.591	0.591	0.635
WormsTwoClass	**0.796**	0.768	0.773	0.779
Average of Accuracy	**0.806**	0.783	0.788	0.795
Average of Ranking	**1.857**	2.857	2.429	2.071
Number of Best Accuracy	**7**	4	3	4

TABLE VIII: Comparison of classification accuracy for each model in the "Sensor" category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
Car	**0.933**	**0.933**	0.883	0.900
Chlorine Concentration	0.816	**0.826**	0.823	0.813
CinC_ECG_torso	0.903	0.871	0.946	**0.959**
Earthquakes	**0.826**	0.823	0.820	0.823
FordA	0.933	0.951	**0.955**	**0.955**
FordB	0.929	0.934	0.934	**0.939**
InsectWingbeat Sound	0.584	0.605	0.625	**0.633**
ItalyPower Demand	0.968	**0.970**	0.959	**0.970**
Lighting2	**0.885**	0.820	0.836	0.836
Lighting7	**0.877**	0.822	0.795	0.849
MoteStrain	**0.895**	0.886	0.855	0.872
Phoneme	**0.363**	0.309	0.306	0.328
Plane	**1.000**	**1.000**	**1.000**	**1.000**
SonyAIBORobot Surface	**0.982**	0.845	0.885	0.870
SonyAIBORobot SurfaceII	**0.940**	0.937	0.903	0.920
StarLightCurves	0.967	0.979	0.979	**0.980**
Trace	**1.000**	**1.000**	**1.000**	**1.000**
wafer	**0.999**	**0.999**	**0.999**	**0.999**
Average of Accuracy	**0.878**	0.862	0.861	0.869
Average of Ranking	2.000	2.222	2.556	**1.833**
Number of Best Accuracy	**11**	6	4	9

TABLE IX: Comparison of classification accuracy for each model in the "Simulated" category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
CBF	0.996	**1.000**	0.982	0.998
MALLAT	**0.981**	0.966	0.956	0.965
ShapeletSim	**0.983**	0.861	0.917	**0.983**
synthetic_control	0.993	**1.000**	**1.000**	**1.000**
Two_Patterns	**1.000**	**1.000**	**1.000**	**1.000**
Average of Accuracy	**0.991**	0.965	0.971	0.989
Average of Ranking	2.000	1.800	2.600	**1.600**
Number of Best Accuracy	**3**	**3**	2	**3**

TABLE X: Comparison of classification accuracy for each model in the " Spectro " category

Dataset	MC-MHLF	LITE	MC-MHLITE	PMC-MHLITE
Beef	**0.933**	0.767	0.833	0.867
Coffee	**1.000**	**1.000**	**1.000**	**1.000**
Ham	0.819	**0.838**	0.743	0.752
Meat	0.983	**1.000**	0.917	0.900
OliveOil	**0.967**	0.900	0.933	**0.967**
Strawberry	**0.980**	**0.980**	0.979	**0.980**
Wine	**0.907**	0.815	0.796	0.815
Average of Accuracy	**0.941**	0.900	0.886	0.897
Average of Ranking	**1.286**	2.000	3.143	2.000
Number of Best Accuracy	**5**	4	1	3